

Incident Response Playbook

Security Operations — Writeup & Reference Document

This document describes the incident response process implemented in the IR Playbook Dashboard project. It covers incident classification, severity levels, detection methodology using Splunk SIEM, per-scenario response playbooks, and lessons learned.

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1. Overview

Security incidents are inevitable in any organisation. Without a structured response process, teams waste time, miss containment windows, and fail to collect forensic evidence. This playbook defines a repeatable workflow for identifying, containing, and mitigating common security threats.

The workflow follows four phases:

- Detection — alerts surface via SIEM monitoring (Splunk)
 - Triage — analyst classifies severity and opens an incident
 - Containment & Response — structured playbook steps executed per incident type
 - Resolution & Review — incident closed, lessons documented
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2. Severity Classification

Every alert is classified into one of four severity levels before a playbook is invoked.

Severity	Criteria	Response SLA
Critical	Active breach, ransomware, data exfiltration in progress	Immediate — 15 min
High	Confirmed attack attempt, phishing with click, brute force threshold exceeded	30 minutes
Medium	Suspicious activity, anomalous traffic, unconfirmed access attempt	2 hours
Low	Informational, failed attempt with no follow-on activity	Next business day

3. SIEM Detection with Splunk

Splunk Enterprise Search & Reporting was used to identify Windows Security events that informed the alert data in this project. The following SPL queries were used:

3.1 Broad event search

```
index="winlog_clients"
```

Returned 2,835 events across Application, Setup, and Security log sources for the period September 2022. Used to establish a baseline of host activity on DESKTOP-4L9QKFS.

3.2 Failed admin logon search

```
source="WinEventLog:*" index="winlog_clients" EventCode=4625 AND Nom_du_compte=Admin*
```

Returned 44 events of failed logon attempts (EventCode 4625) for admin accounts over a 24-hour window. This pattern is consistent with a brute force login attack. The results were saved as a Splunk report: "WINDOWS - Connections failed for admin account".

EventCode	Description	Mapped Alert Type
4625	Failed logon attempt	Brute Force Login Attempt
4624	Successful logon	Unauthorized Access Attempt
4634	Account logoff	Unauthorized Access Attempt
4672	Special privileges assigned	Unauthorized Access Attempt
4799	Security-enabled group queried	Unauthorized Access Attempt

Screenshots of these Splunk searches are included in the /screenshots directory of the project repository.

4. Incident Types & Response Playbooks

The following playbooks are implemented in the dashboard. Each playbook is invoked when an analyst opens an alert of the corresponding type.

4.1 Ransomware Signature Detected

Severity: Critical

Response Steps:

- 1. Immediately isolate the affected endpoint from the network
- 2. Identify the ransomware variant using AV or threat intel feeds
- 3. Preserve forensic evidence — capture memory dump and disk image
- 4. Notify security lead and initiate incident escalation
- 5. Determine blast radius — identify other potentially infected hosts
- 6. Restore affected systems from clean, verified backups
- 7. Patch the initial attack vector before reconnecting to network
- 8. Conduct post-incident review and update playbook

4.2 Unusual Data Exfiltration

Severity: Critical

Response Steps:

- 1. Identify the user account and endpoint involved

- 2. Block outbound traffic from the source IP immediately
- 3. Determine what data was accessed and volume transferred
- 4. Preserve logs from DLP, proxy, and endpoint telemetry
- 5. Suspend the user account pending investigation
- 6. Notify legal, compliance, and data protection officer
- 7. Assess whether breach notification obligations apply
- 8. Review and tighten DLP policies post-incident

4.3 Phishing Email Detected

Severity: High

Response Steps:

- 1. Identify all recipients of the phishing email
- 2. Pull and block the sender domain and IP at the email gateway
- 3. Check if any recipient clicked links or opened attachments
- 4. Reset credentials for any accounts that interacted with the email
- 5. Scan endpoints of affected users for malware indicators
- 6. Review mailbox activity for forwarding rules or suspicious logins
- 7. Send user awareness notification across the organisation

4.4 Brute Force Login Attempt

Severity: High

SIEM Detection: Detected via Splunk query on EventCode 4625 — 44 failed admin logon events identified on host DESKTOP-4L9QKFS within a 24-hour window.

Response Steps:

- 1. Identify the targeted account and source IP address
- 2. Lock the account if the login threshold has been exceeded
- 3. Block the source IP at the firewall or WAF
- 4. Review authentication logs for any successful logins from that IP
- 5. Enable or enforce MFA on the targeted account
- 6. Check for lateral movement if account was compromised
- 7. Tune IDS/IPS rules to alert on similar patterns going forward

4.5 DDoS Traffic Spike

Severity: Medium

Response Steps:

- 1. Confirm the spike is malicious and not a legitimate traffic surge
- 2. Enable rate limiting at the edge or CDN layer
- 3. Contact upstream ISP or DDoS mitigation provider for scrubbing
- 4. Identify source IPs or ASNs and apply null routing where possible
- 5. Monitor service availability and escalate if SLAs are breached
- 6. Capture traffic samples for post-incident analysis
- 7. Review and update DDoS response runbooks

4.6 Malware Signature Detected

Severity: Critical

Response Steps:

- 1. Isolate the affected host from the network immediately
- 2. Run a full antivirus and EDR scan on the endpoint
- 3. Collect forensic artefacts — process list, network connections, prefetch files
- 4. Identify the malware family and check for known indicators of compromise
- 5. Check for persistence mechanisms (registry keys, scheduled tasks, startup items)
- 6. Determine if the malware has spread to other hosts
- 7. Restore from a known-good backup after full remediation

4.7 Unauthorized Access Attempt

Severity: Medium

SIEM Detection: Detected via Splunk — EventCodes 4624 (successful logon), 4634 (logoff), and 4672 (special privileges assigned) observed within the winlog_clients index.

Response Steps:

- 1. Review access logs to determine what resources were targeted
- 2. Verify if the attempt succeeded or was blocked
- 3. Identify the source IP and correlate with threat intelligence
- 4. Check for other attempts from the same IP or user agent
- 5. Strengthen access controls or MFA on targeted resources
- 6. Document findings and close the incident with a timeline

5. Incident Status Workflow

Each incident progresses through the following statuses in the dashboard:

Status	Meaning	Who Sets It
Open	Alert detected, not yet assigned	System / SIEM
Investigating	Analyst actively working the incident	Analyst
Contained	Threat neutralised, evidence being gathered	Analyst / Lead
Resolved	Root cause identified, systems restored	Security Lead

6. Lessons Learned

The following observations were made during the development of this project and the analysis of Splunk event data:

- EventCode 4625 bursts (44 events in 24 hours targeting admin accounts) are a reliable brute force indicator and should trigger automatic account lockout policies if not already configured.
 - Separating alert data from playbook logic (database vs. static dict) makes the system easier to extend — new alert types can be added to the DB without touching the playbook code.
 - A severity filter alone is insufficient for analyst workflow — future versions should include status-based filtering and assignment to specific analysts.
 - Static HTML dashboards are viable for demos but limit real-time updates; a production SOC tool would use WebSockets or periodic polling for live alert feeds.
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7. References

- Splunk Enterprise Search & Reporting — <https://www.splunk.com>
- Windows Security Event Log EventCodes — Microsoft Documentation
- NIST SP 800-61 Rev 2: Computer Security Incident Handling Guide
- SANS Incident Handler's Handbook

End of document.